



# Do Space Rovers Promise A Better Future For Space Technology?

In the rapidly transforming technological world, India is currently focusing its attention on emerging technologies. Though not by design, space rover technology is still an emerging sector in India



Chandrayaan-2 Vikram lander and rover before launch (Credit: Indian Space Research Organisation)



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A space rover is sometimes called a planetary rover. It is a space exploration vehicle designed to move along the surface of a planet or other celestial body. Some space rovers are autonomous robots.

Rovers are usually sent to a planet using a lander-style spacecraft. These are used to gather various information about the planet or take samples and/or pictures. These help in the exploration of the universe.

With abundant technologies, space also offers opportunities for rover technology. Space rovers are increasingly being seen in the field of space technology. These are becoming essential for space explora-

tion because of better service compared to satellites. Today, rovers contain several instruments for overall improvements in exploration and findings in planets, sub-planets and other celestial bodies.

A rover contains the following parts:

- Body, which is the structure that protects the rover's vital organs.
- Brains, which are computers that process various information.
- Temperature controls, which include internal heaters, a layer of insulation and more.
- Neck and head, which are the mast for cameras to give the rover a human-scale view.
- Eyes and other senses, which include cameras and other instruments to give the rover information about the environment around it.
- Arm and hand, which help extend the rover's reach and collect rock samples for study.
- Wheels and legs, which provide mobility for the rover.
- Energy, which includes batteries and power control units.
- Communications, which includes antennae for speaking and listening.

## Space rovers giving new look to lunar missions

After manned missions to the moon, the lunar space sector undertook several experiments. Establishing a hotel in space in the future is a dream of space scientists. The lunar space sector cannot survive without innovations and responsible handling by space scientists who have specialised